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NPS/003/016 – Technical Specification for 48V and 110V Battery and Charger Systems

1. Purpose

This document is the technical specification for 48V and 110V battery and charger systems for use on the NEDL and YEDL networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title

2. Scope

This specification details the requirements for 48V and 110V battery and charger systems for use at NEDL and YEDL substations.

3. Technical Requirements

110V and 48V battery systems shall be in accordance with IEC 60623, ENATS 50-18, and Appendices 1 and 3 of this document.

The technical specification of any system must be confirmed using the table in Appendix 2.

The project specific requirements including the required battery Discharge Profile are specified in Appendix 3.

4. References

4.1 External Documentation

Reference	Title
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4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	10/02/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	10/02/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D.Gazda	Standards Team Co-ordinator	p.p. Dave Blackledge	14/02/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	24/03/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	15/02/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	20/02/2006

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Appendix 1: Technical Specification for 48V and 110V Battery Systems

	ALL SYSTEMS	110V	48V
System Output Characteristics			
Nominal voltage		108V	46.8V
Float voltage (max acceptable)		127.8V	55.38V
Minimum voltage after specified discharge profile		102V	44.5V
Maximum charger voltage under all conditions		137.5V	62.5V
Battery			
Standard	BS EN 60623		
Type	Low maintenance gas-recombination Nickel Cadmium		
Design life	20 years		
Battery size	To meet the discharge profile specified in Appendix 3.		
Ageing factor	1.1		
Minimum topping-up period	10 years		
Minimum number of cells in battery		90	39
Maximum float voltage per cell		1.42V	1.42V
Rated temperature	15°C		
Temperature range	0°C – 40°C		
Mounting	Cubicle		
Connectors	Shrouded		
Isolation facilities required	From charger and distribution board		
Battery earthing	The battery shall be unearthed		
Charger Supply			
AC Supply	240V AC single phase or 415V AC 3 phase.		
Nominal supply frequency	50Hz		
Supply frequency range	47-52Hz		
Charger			
Type	Constant voltage, two-stage, temperature compensated.		
Ambient temperature range	0°C - 40°C		
Maximum Float Voltage		127.8V	55.38V
Maximum voltage under all conditions		137.5V	60V
Voltage adjustment facility	Float and boost voltages should be adjustable however such controls should not be on the front facia.		
Rated charging current	To cover standing load as specified in Appendix 3 plus adequate charging current		
Normal float charge	50% rated load at rated input voltage.		
Variation in charging voltage as input voltage varies over range of -10% + 6% of rated value:			

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Across load range 0 – 10%	Maximum 2% variation in output voltage		
Across load range 10% - Full Load	Maximum 1% variation in output voltage		
Variation in current limit over the specified voltage range of the battery.	Maximum 2% of the nominal setting		
Frequency range over which charger performance should be met.	47-52Hz		
Max earth current with one pole of battery earthed (at maximum battery voltage)	5mA		
Transformer screening	Earth metal screen between primary and secondary windings.		
System Monitoring			
High voltage alarm	Local indication plus spare contact	132V	57.6V
Low voltage alarm	Local indication plus spare contact	117V	51V
Charger fail alarm	Local indication plus spare contact		
Battery earth fault alarm	Local indication plus spare contact	50kΩ	50kΩ
Battery high resistance or open circuit	Local indication plus spare contact		
Monitoring alarm protection	2A HRC fuse		
Battery earth fault alarm test facility			
Test resistor value	10% below sensitivity of the earth fault relay		
Operating switch	A “return-to-off” on/off switch		
Discharge Test Facility			
Discharge resistor	To give one hour discharge rate		
Operating method	Via a “return-to-off” on/off switch, and contactor.		
Warning label	“This switch shall not be operated for longer than 10 seconds”		
Instruments			
Battery output voltmeter	Push-button or self-return-switch operated		
Charger output ammeter			
Cubicles and Battery Stands			
Design	Generally as drawing Y042S0900		
Design life	40 years		
Cubicle cable entry	Bottom		
Cubicle Colour	Light grey		
Exposed conductors	Shrouded		
Doors	Lockable		
Distribution Board			
General arrangement	As drawing Y042S0900, 901, 902 and 903		
Number of ways	As specified in Appendix 3		
Rating of ways	16A or 32A as specified in Appendix 3		
Labels	As specified in Y042S0900, 901, 902 and 903		

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Fuses			
Type	Cartridge type: BS88		

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Appendix 2: Declaration of technical specification of 48V and 110V Battery Systems

	UNIT	ALL SYSTEMS	110V	48V
System Output Characteristics				
Nominal voltage	V			
Float voltage (max acceptable)	V			
Minimum voltage after specified discharge profile	V			
Maximum charger voltage under all conditions	V			
Battery				
Standard	BS/IEC?			
Type of cells				
Design life	Years			
Aging factor applied				
Minimum topping-up period	Years			
Number of cells in battery	No.			
Float voltage per cell	V			
Rated temperature	°C			
Temperature range	°C			
Cell mounting: Cubicle? Tiered?	State			
Connectors: Shrouded?				
Isolation from both dist board and charger?				
Battery earthing: Unearthed?				
Charger Supply				
AC Supply	V + θ			
Nominal supply frequency	Hz			
Supply frequency range	Hz-Hz			
Charger				
Type				
Ambient temperature range	°C - °C			
Maximum Float Voltage	V			
Boost voltage	V			
Voltage adjustment facility: Float and boost	Y/N			
Rated charging current	A			
Normal float charge	A			
Variation in charging voltage as input voltage varies over range of -10% + 6% of rated value:				
Across load range 0 – 10%	%			
Across load range 10% - Full Load	%			
Variation in current limit over the specified voltage range of the battery.	%			

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Frequency range over which charger performance should be met.	Hz - Hz			
Max earth current with one pole of battery earthed (at maximum battery voltage)	mA			
Transformer screened as per specification?	Y/N			
System Monitoring				
High voltage alarm setting	V			
Low voltage alarm setting	V			
Charger fail alarm	V			
Battery earth fault alarm	kΩ			
Battery high resistance or open circuit	Yes/No			
All alarms have local indication?	Yes/No			
Contacts for remote alarm: Individual/Common?				
Alarm circuit fuse rating	A			
Battery earth fault alarm test facility				
Is this provided?	Yes/No			
% rating against E/F relay sensitivity	%			
Discharge Test Facility				
Is this provided?	Yes/No			
Discharge rate				
Operating method				
Warning label provided?	Yes/No			
Instruments				
Battery output voltmeter	Yes/No			
Charger output ammeter	Yes/No			
Cubicles and Battery Stands				
Design life	Years			
Cubicle cable entry position				
Cubicle Colour				
Exposed conductors shrouded	Yes/No			
Doors lockable	Yes/No			
Distribution Board				
Number of ways				
Rating of ways				
Fuses				
Type				
AC interrupting current	kA			

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Appendix 3: Addendum to supplier requirements

Project specific installation and protection requirements will be provided by Primary Engineering Projects for inclusion in this appendix.